Lab: File Management & Version Control

Linux Basics & Git Basics

Scenario

You have been given a CSV file with student data. Organize it properly and push to GitHub using version control.

Task 1: Create Directory Structure

Create a project folder with subdirectories to organize your files.

```
mkdir my_project
cd my_project
mkdir data docs
```

What to do:

1. Create a main directory
2. Create two subdirectories: `data/` and `docs/`
3. Verify structure using `ls -la`

Task 2: Copy and Organize Data

Copy the provided `student_data.csv` into the `data/` folder.

```
cp student_data.csv data/
ls -la data/
cat data/student_data.csv
```

What to do:

1. Copy the CSV file to `data/` folder
2. Verify the file is there
3. View the contents of the CSV file

Task 3: Create README File

Create a README.md file explaining your project.

```
cat > README.md << 'EOF'
# Student Data Management

A simple system to organize student information.

## Files
- data/student_data.csv: Contains student records
- docs/: Documentation files
EOF

cat README.md
```

What to do:

1. Create a README.md file
2. Add project description
3. Explain what the project contains

Task 4: Create Documentation

Create a file explaining the CSV structure.

```
cat > docs/README.txt << 'EOF'
Student Data Format

Columns in student_data.csv:
- RollNo: Student ID
- Name: Student Name
- Department: Course/Department
- CGPA: Academic Grade

Total Records: 10 students
EOF

cat docs/README.txt
```

What to do:

1. Create documentation in `docs/` folder
2. Explain what each column means
3. Describe your data

Task 5: Initialize Git

Set up version control for your project.

```
git init
git config --global user.name "Your␣Name"
git config --global user.email "your@email.com"
git status
```

What to do:

1. Initialize git repository
2. Configure your name and email
3. Check status with `git status`

Task 6: Commit Files

Stage and commit your files to version control.

```
git add .
git status
git commit -m "Add␣student␣data␣and␣documentation"
git log
```

What to do:

1. Stage all files using `git add .`
2. Check status
3. Create a commit with message
4. View commit history

Task 7: Push to GitHub

Upload your project to GitHub.

```
git remote add origin https://github.com/USERNAME/REPO.git
git branch -M main
git push -u origin main
```

What to do:

1. Create a new PUBLIC repository on GitHub
2. Copy the repository URL
3. Connect your local repo to GitHub
4. Push your code to GitHub
5. Verify files are on GitHub

Linux Commands Reference

Basic Commands

- `mkdir`: Create directory
- `cd`: Change directory
- `ls`: List files
- `ls -la`: List files with details
- `cp`: Copy files
- `cat`: View file contents
- `pwd`: Print current directory
- `touch`: Create empty file

Git Commands Reference

Git Commands

- `git init`: Initialize repository
- `git config --global`: Set user info
- `git add .`: Stage all files
- `git status`: Check status
- `git commit -m`: Save changes
- `git log`: View history
- `git remote add`: Connect to GitHub
- `git push`: Upload to GitHub

Expected Structure

```
my_project/
|-- README.md
|-- data/
|   |-- student_data.csv
‘-- docs/
    ‘-- README.txt
```

Submission Checklist

Before Submitting

- Directory structure created
- CSV file copied to data/ folder
- README.md created in main directory
- Documentation created in docs/ folder
- Git initialized
- All files committed
- GitHub repository created (PUBLIC)
- Code pushed to GitHub
- GitHub link provided

Questions

1. What is the purpose of `git init`?
2. What does `git add .` do?
3. What is a commit?
4. Why use GitHub?
5. How do you view commit history?

Duration: 1 Hour — Difficulty: Beginner